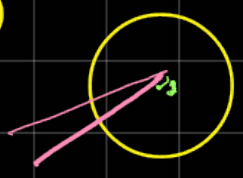
External memory magnetic Disk ⇒ maths → Avg value can't be used.

Avg time for track 1 & track 2.

1)



→ made of glass.

→ Rotation → has dynamic/momentum

↳ glass is resistant to it

lower fly height

Distance between head & surface.



1 read = 1 byte stored

more capacity

faster

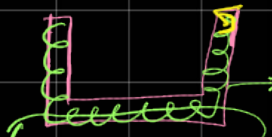
less space to store some data.

A1 < A2

↳ store more data

more glass substrate us

⇄ 2 direction of current.

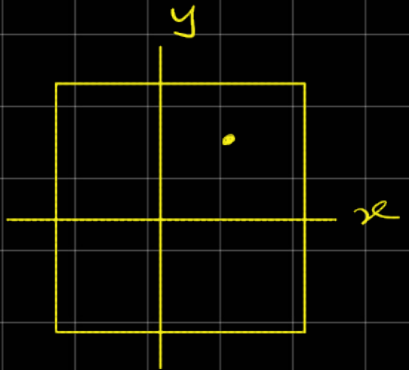


→ head made of coil → use EMF.

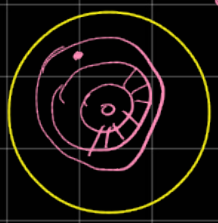
→ magnetizable surface.

→ write operation.

Locate a particular point on the surface:



sectors & tracks

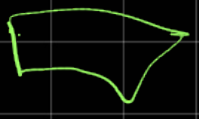


↓

concentric ring across the surface.

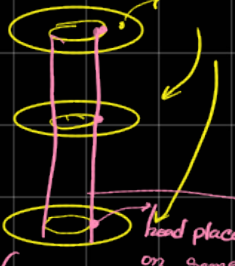
if i move head across the surface → misAligned with the adjacent track so gap maintain.

↳ inter-track gap.

outer track sector bigger 

inner " " small 

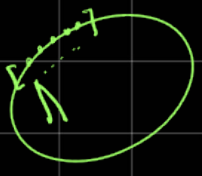
both upper & lower surface → store data.

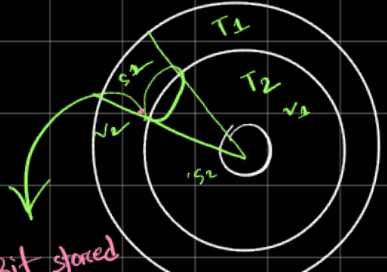


one single data accessed simultaneously across multiple disk.

1 bit → 3×2 = 6 bits → single position → 8 bit at a time.

↓ both sid





access S1 from tra

bit near th

hence with faster speed $v_1 = \frac{d_1}{t}$ $v_2 \uparrow = \frac{d_2 \uparrow}{t}$

$v_2 > v_1$

s_1, s_2
Same number of bits

Store angular momentum, $\omega_1 = \frac{\theta_1}{t}$, $\omega_2 = \frac{\theta_2}{t}$

$s_1 \rightarrow$ bit density higher

$s_2 \rightarrow$ wastes space.

Same but linear velocity different.

s_1, s_2 store same no. of bits.

bit density const
↳ # of bits / length same.

Different zone → different sector size.

1st one → const angular velocity (CAV)

2nd one → multiple zone Recording (MZR)

→ 2 pics of book. → 198 page

Bad sector
↳ permanently damaged.

1TB Storage → some for operational purpose

600 byte/sector
size of a single sector

17 in the beginning, 20 at the end

Each track should have their own head.

↳ flexible floppy disk.

Head Mechanism
↳ how it is placed.
connected → fixed gap head

floppy → Contact head

light paper?
lowest fly height
zero fly height but during rotation, air pressure will uplift the coil / head.
↳ Aerodynamic.
→ অসুস্থ (SICK!)

seek time :
↳ time to locate track.
1.
2. Rotate
↳ sector selection time.

Desired disk busy → serve request using → ?

System bus shared → get the access of system bus.
↳ wait for it.

ignoring waiting time.

$$T = T_s + \frac{1}{2r} + \frac{b}{rN}$$

T_s ↓ seek
r ↓ rotational speed
b ↓ no of bits
rN ↓ Transfer time

2 portion (the 1 with less time will be selected)

$$T = T_s + \frac{1}{2r_0} + \frac{1}{r_0}\frac{b}{N}$$